

# ScAN - A Path to PIM Hardware-aware Network

## Student Network

| Layer | Operation         | Cin -> Cout | Kernel | Stride | Output size        |
|-------|-------------------|-------------|--------|--------|--------------------|
| 1.1   | Conv+BN+ReLU      | 4 -> 96     | 3x3    | 1      | 96 x 16 x 16       |
| 1.2   | Conv+BN+ReLU      | 96 -> 96    | 3x3    | 1      | 96 x 16 x 16       |
| 1.3   | Conv+BN+ReLU      | 96 -> 96    | 3x3    | 2      | 96 x 8 x 8         |
| 2.1   | Conv+BN+ReLU      | 96 -> 192   | 3x3    | 1      | 192 x 8 x 8        |
| 2.2   | Conv+BN+ReLU      | 192 -> 192  | 3x3    | 1      | 192 x 8 x 8        |
| 2.3   | Conv+BN+ReLU      | 192 -> 192  | 3x3    | 2      | 192 x 4 x 4        |
| 3     | Conv+BN+ReLU      | 192 -> 192  | 3x3    | 1      | 192 x 4 x 4        |
| 4     | Conv+BN+ReLU      | 192 -> 192  | 1x1    | 1      | 192 x 4 x 4        |
| 5     | Conv logits + GAP | 192 -> 100  | 1x1    | 1      | 100 x 4 x 4 -> 100 |

## Simulator

All reported quantized accuracies use the fixed simulator on CIFAR-100 RGGGB with raw\_noise=True (50k train / 10k test). The dense int4 path uses unsigned I16 first-layer input codes in [0,65535], integer W4/LUT weights, unsigned A4 hidden activations in [0,15], int32 convolution, and precomputed requantization. BatchNorm is folded before simulator execution. Runtime forward does not run tanh, k-means, nearest-LUT search, or weight quantization.

Current LUT: 15 integer entries, [-14,-12,-10,-8,-6,-4,-2,0,2,4,6,8,10,12,14] with m=15.

## QKD Results

| Stage | Checkpoint      | Train best | Simulator | Notes                                            |
|-------|-----------------|------------|-----------|--------------------------------------------------|
| SS    | ..._lutq_ss.pth | 60.66%     | 60.21%    | Self-studying, I16/W4/A4 LUT-Q + PACT.           |
| CS    | ..._lutq_cs.pth | 61.39%     | 61.24%    | Co-studying; improvement appeared after LR drop. |
| TU    | ..._lutq_tu.pth | 61.81%     | 61.66%    | Tutoring; best current int4 simulator result.    |

## PIM Conv Status

Implemented an explicit PIM-QAT model path and PIM simulator path without changing the default dense-QKD path. The current minimal PIM path keeps the existing BN student model; GBN/PCN are not enabled in this pass.

The PIM path follows pim\_info/scanpcn\_inference\_spec.tex for row select over [18,18,36,72] chunks, keep 48 of 144 rows by default, hybrid stage1.1 keep 72 of 144 rows, 7-bit mid-rise tile ADC, digital BN scale/bias, normalized hidden DAC activations in [0,1], and unmasked stage1.0 / logits layers.

PIM-QAT eval and PIM simulator are numerically matched on the TU checkpoint smoke input: max\_abs\_diff <= 6e-7. PIM-specific training is the next experiment and is kept separate from the dense int4 QKD results above.

## Current Best

checkpoints/rerun/qkd\_i16\_intlut\_pact\_g8m15\_lutq\_tu.pth